PREPARATION FOR ANESTHESIA

I. INDUCTION OF ANEASTHESIA
II. TECHNIQUE OF OROTRACHEAL INTUBATION
III. COMPLICATION OF TRACHEAL INTUBATION


Definition of general anesthesia

1. Amnesia
2. Sedation/Hypnosis
3. Analgesia
4. Immobility
5. Suppressed Autonomic Reflexes

Principles of anesthesia

- ✓ Anesthesia defined as the abolition of sensation
- ✓ Analgesia defined as the abolition of pain

- ✓ "Triad of General Anesthesia"
  - ✓ need for unconsciousness
  - ✓ need for analgesia
  - ✓ need for muscle relaxation

Pre anesthetic evaluation

- ***Pre anesthetic evaluation*/ The preoperative evaluation** is evaluation of the patients medical physical and mental status before surgery.

- **It** serves as a screening tool to anticipate and avoid problems.

- **The goals of a preoperative evaluation** are to reduce patient risk and morbidity associated with surgery and coexisting diseases, promote efficiency and reduce costs, as well as to prepare the patient medically and psychologically for surgery and anesthesia .

A good anesthetic begins with a good plan.

- Psychological support and pre medication provided by the anesthetist
- ***Pre medication*** -administration of drugs orally ,Iv or IM in the period of 1-2 hours before induction of anesthesia .
- For out patient surgery premedication may be administered intravenously in the immediate preoperative period.
- Preoperative visit and interview patient and family.
- A thorough description of planned anesthetic and events anticipated in the pre operative period serves as a non pharmacologic antidote to anxiety.

Preparation for anesthesia

- Most patients scheduled for surgery will develop some degree of apprehension .
- The psychological stress a patient experiences prior to surgery can be more than the actual physical insult

Some of the causes of pre operative anxiety;

1.The fear of separation from family.

2.The fear of dying during operation.

3.The inability of preserving modesty during the operation.

4.The fear of experiencing pain after the operation.

5.The fear of surgical mutilation and an altered body image.

Various Goals for Preoperative Medication

- Relief of anxiety
- Sedation
- Amnesia
- Analgesia
- Drying of airway secretions
- Prevention of autonomic reflex responses
- Reduction of gastric fluid volume and increased pH
- Antiemetic effects
- Reduction of anesthetic requirements
- Facilitation of smooth induction of anesthesia
- Prophylaxis against allergic reactions

General contraindications to the use of a premedication include:

1.Allergy or hypersensitivity to the drug.

2. Upper airway compromise, or respiratory failure.

3. Hemodynamic instability or shock.

4. Decreased level of consciousness.

5. Severe liver, renal, or thyroid disease.

6. Obstetrical patients.

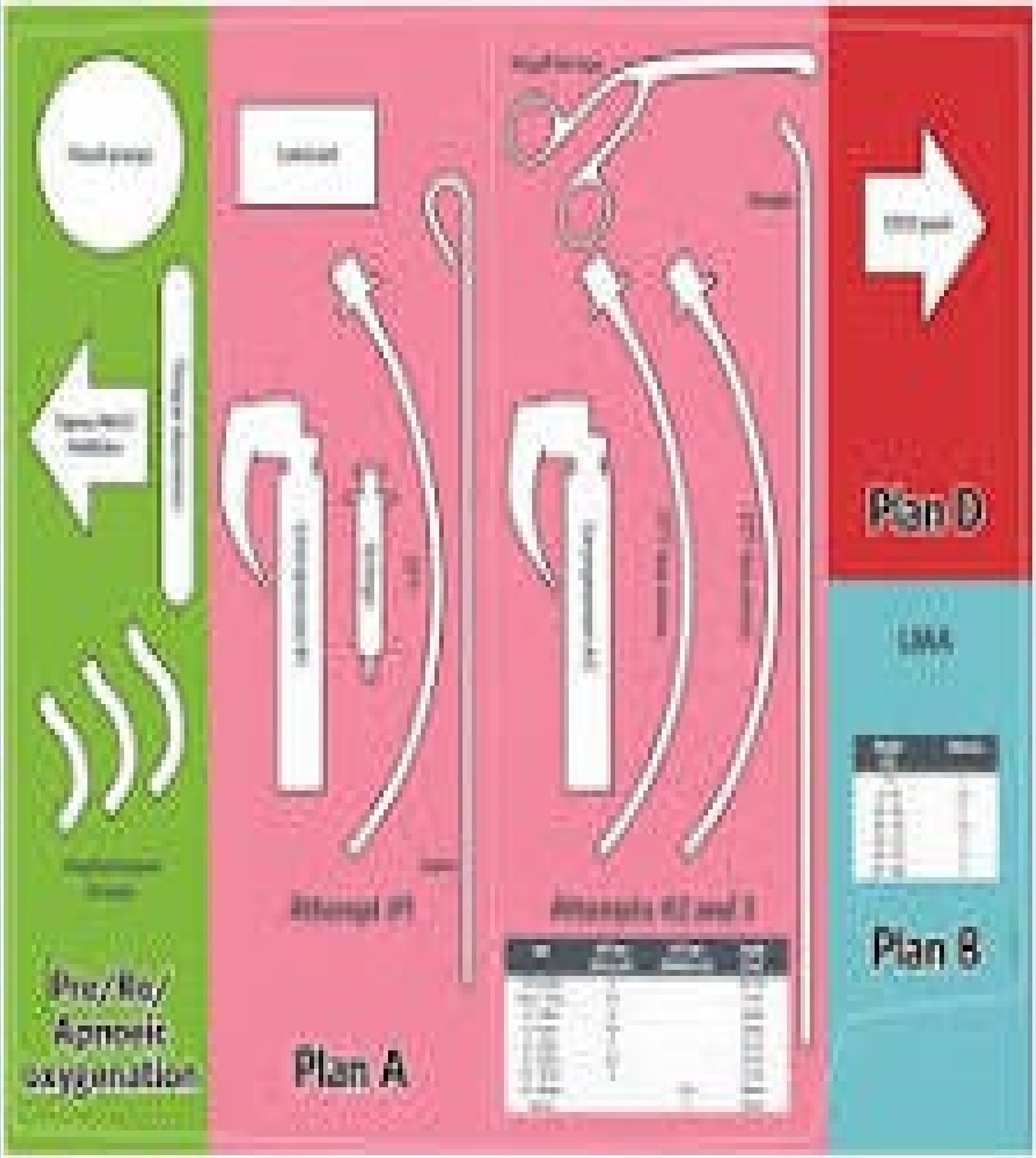
7. Elderly or debilitated patients

Ideal Anesthetic

- Immediate onset
- Reversible
- Appropriate duration
- No permanent damage
- No tissue irritation / pain
- Wide therapeutic range
- Effective regardless of application

Routine preparation before induction of anesthesia

- SAM (SAMMM') stands for:
- **S** Suction checked and functioning.
- **A** Airway equipment checked and prepared. (This includes checking that you have a functioning and backup laryngoscope, an appropriate sized endotracheal tube and stylet, oropharyngeal airways, as well as an oxygen source and manual resuscitation bag).
- **M** Machine checked.(Attach an anesthetic breathing system with properly sized face mask and check breathing system valves ,caliber the oxygen analyzer with air and oxygen and set alarm, check the carbon dioxide absorbent for color change ,check the liquid level of vaporizer ,check the function of ventilator ,and check the final position of all flow meter, vaporizer and monitors.(visual and audible alarm))
- **M** Monitors available and functioning.
- **M** Medications prepared and labeled. You should know where the emergency drugs are kept and location of the difficult intubation cart.


Plan A
Plan D
LMA
Plan B

Monitors

- Blood pressure apparatus Bladder length should be sufficient to encircle at least 80% of the extremity.
- Falsely high estimates result when cuffs are too small, when cuffs are applied too loosely, or when the extremity is below heart level.
- Falsely low estimates result when cuffs are too large, when the extremity is above heart level, or after quick deflations.
- Pulse oximetry ;Pulse oximetry is a noninvasive method by which arterial oxygenation can be approximated.
- Saturation as measured by pulse oximetry is denoted by SpO_2.
- Electrocardiography;
- capnograph ;Capnography is the continuous monitoring of a patient's capnogram.
- A capnogram is a continuous concentration–time display of the CO_2 concentration sampled at a patient's airway during ventilation.

Pre-induction Preparation and necessary equipment

- Apply monitors (make sure they are working properly) &obtain baseline vitals
- Position to the patient for optimal airway management
- Pre-medicate if indicated (anxiolysis/amnestic, anti-sialagogue, vagal reflex prophylaxis

Equipment

- Endotracheal tube with different size
- Air way
- Suction catheter
- Oral and nasal air way
- Laryngeal mask air way
- Nasogastric tube
- Temperature probe
- Iv solution ,IV cannula

Endotracheal tube recommendations

A. Endotracheal tube size (mm): for children older than 2 years ETT can be estimated by: Age/4 + 4.

B. Length of Insertion (cm) of ETT

1. Under 1 year: 6 + Wt(kg).
2. Over 2 years: 12 + Age/2.
3. Multiply internal diameter (mm) of ETT by 3 to give insertion (cm).
4. Add 2-3 cm for nasal tube.

C. Pediatrics: generally use uncuffed tubes in patients under 10 years. When a cuff tube is used maintain endotracheal leak at 15-20 cm H_2O.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Induction of anesthesia

- Induction of anesthesia produces an unconscious patient with depressed reflexes who is entirely dependent on the anesthetist for maintenance of homeostatic mechanisms and safety.

- The patient's position for induction and intubation is usually supine, with extremities resting comfortably on padded surfaces in a neutral anatomic position.

- The head should rest comfortably on a firm support

- Routine pre induction administration of oxygen minimizes the risk of hypoxia developing during induction of anesthesia.

- High flow (8 to 10 l/min) oxygen should be delivered via a face mask placed gently on the patient's face.

- The patient can be instructed to take deep breaths and exhale fully to speed the exchange of oxygen.

WHY ARE WE INDUCING?

- To facilitate tolerance of airway management
- To facilitate tolerance of surgery (the transition from induction to maintenance phase

WHICH DRUGS DO WE USE?

- ✓ Based upon method of **airway management** (ETT vs. LMA
- ✓ Based upon available **routes of administration** (is there an IV?)
- ✓ Based upon drug **side effects**
- ✓ Based upon **patient** (age, vitals, disease states – i.e. heart, liver, or kidney failure, etc.)
- ✓ Based upon **clinical scenario** (indication for RSI vs. standard induction, emergency, expected difficult airway)
- ✓ Based upon type of **surgery** (duration, need for paralysis, etc.)

CNS Depression

- **Sedative/Hypnotics (IV anesthetics)**
  - Propofol (1.5-2.5mg/kg)
  - Etomidate (0.2-0.5mg/kg)
  - Ketamine (1-2mg/kg)
- **Barbiturates**
  - Thiopental (3-6mg/kg)
  - Methohexital (1-2mg/kg)
- **Benzodiazipines**
  - Midazolam (Sedation:0.01-0.1mg/kg; Induction:0.1-0.4mg/kg)
  - Diazepam (S:0.04-0.2mg/kg; I:0.3-0.6mg/kg)
- **Inhaled Anesthetics**
  - Halogenated agents
  - N2O

RESPIRATORY REFLEX DEPRESSION

- **Opiod Narcotics**
  - Fentanyl (1-2mcg/kg)
  - Morphine (0.1-1mg/kg)
  - Meperidine (poor analgesic, 2 – 5 mg/kg)
- **Local Anesthetics**
  - Lidocaine (1-1.5mg/kg)

MSK RELAXATION

- **Depolarizing:**
  - Succinylcholine (1-2mg/kg)
- **Non-Depolarizing:**
  - Rocuronium (0.6-1.2mg/kg),
  - Vecuronium (0.1mg/kg),
  - Pancuronium 0.04-0.1mg/kg],
  - Cisatracurium 0.2mg/kg

Techniques of Direct Laryngoscopy & Intubation

- The choice of induction technique is guided by the patient's medical condition, anticipated airway management (i.e., risk of aspiration, difficult intubation, or compromised airway), and patient preference.

- Inserting tube into the trachea has become a routine part of delivering a general anesthetic.

- ET is not a risk-free procedure, however, and not all patients receiving general anesthesia require ET

- TT is often placed to protect the airway and for airway access

Indications for Intubation

Common indications for endotracheal intubation:

- ✓ Provide patent airway, protection from aspiration
- ✓ Facilitate positive-pressure ventilation
- ✓ Operative position other than supine
- ✓ Operative site near or involving the upper airway,
- ✓ Airway maintenance by mask is difficult, disease involving the upper airway
- ✓ One-lung ventilation
- ✓ Altered level of consciousness, tracheobronchial toilet
- ✓ Severe pulmonary or multisystem injury

Preparation for intubation

- Includes checking equipment and properly positioning the patient.
- The TT should be examined.
- The tube's cuff inflation system can be tested by inflating the cuff using a 10-mL syringe.
- Maintenance of cuff pressure *after detaching the syringe ensures proper cuff and valve function.*
- The connector should be pushed into the tube as far as possible to decrease the likelihood of disconnection.
- This shape facilitates intubation of an anteriorly positioned larynx.
- The desired blade is locked onto the laryngoscope handle, and bulb function is tested.
- The light intensity should remain constant even if the bulb is jiggled

- If a stylet is used, it should be inserted into the ETT, which is then bent to resemble a hockey stick .

- A blinking light signals a poor electrical contact, whereas fading indicates depleted batteries.

- An extra handle, blade, ETT (one size smaller), and stylet should be immediately available.

- A functioning suction unit is needed to clear the airway in case of unexpected secretions, blood, or emesis

- Successful intubation often depends on correct patient positioning. The patient's head should be level with the anesthetists waist or higher to prevent unnecessary back strain during laryngoscopy.

- Rigid laryngoscopy displaces pharyngeal soft tissues to create a direct line of vision from the mouth to the glottic opening
- Moderate head elevation (5–10 cm above the surgical table) and extension of the Atlanta occipital joint
- Place the patient in the desired sniffing position
- The lower portion of the cervical spine is flexed by resting the head on a pillow.

preoxygenation

Preoxygenation with several deep breaths of 100%oxygen provides an extra margin of safety in case the patient is not easily ventilated after induction.

The establishment of a patent airway is probably our most important safety concern.

Disaster overtakes the patient within a matter of minutes if he cannot breathe for himself (because we paralyzed him), and we cannot ventilate his lungs(because his airway is obstructed by soft tissue and because we cannot intubate his trachea for any number of reasons).

Before inducing apnea, we replace the nitrogen in his lungs with oxygen, we can gain 3 to 6 minutes (more with a large functional residual capacity (FRC)) before arterial hypoxemia occurs.


- There fore ,we routinely pre-oxygenate patients before inducing anesthesia.
- This procedure is simple: we apply a face mask and select a flow of oxygen high enough to prevent the patient from inhaling his exhaled nitrogen.
- This period not to injure the patient's eyes by unintentionally abrading the cornea.
- Thus, the eyes are routinely taped shut, often after applying a petroleum-based ophthalmic ointment.

ETT under anesthesia

A. Preoperative evaluation of the airway will help determine the route (oral or nasal) and method (awake or anesthetized) for tracheal intubation.

B. Equipment: laryngoscope with working light, endotracheal tubes of appropriate sizes, malleable stylet, oxygen supply, functioning suction catheter, functioning IV, and appropriate anesthetic drugs

B. Hold the laryngoscope in the palm of the left hand and introduce the blade into the right side of the patient's mouth.

Advance the blade posteriorly and toward the midline, sweeping the tongue to the left.

Check that the lower lip is not caught between the lower incisors and the laryngoscope blade.

The placement of the blade is dependent on the blade used.

1. Macintosh (curve) blade: The tip of the curved blade is advanced into the vallecula (the space between the base of the tongue and the pharyngeal surface of the epiglottis.

Cont…….

2. Miller (straight) blade: The tip of the straight blade is passed beneath the laryngeal surface of the epiglottis, epiglottis is then lifted to expose the vocal cords.

C. Regardless of the blade used, lift the laryngoscope upward and forward, in the direction of the long axis of the handle, to bring the larynx into view.

Do not use the upper incisors as a fulcrum for leverage because this action may damage the upper incisor and may push the larynx out of sight.

Con…….

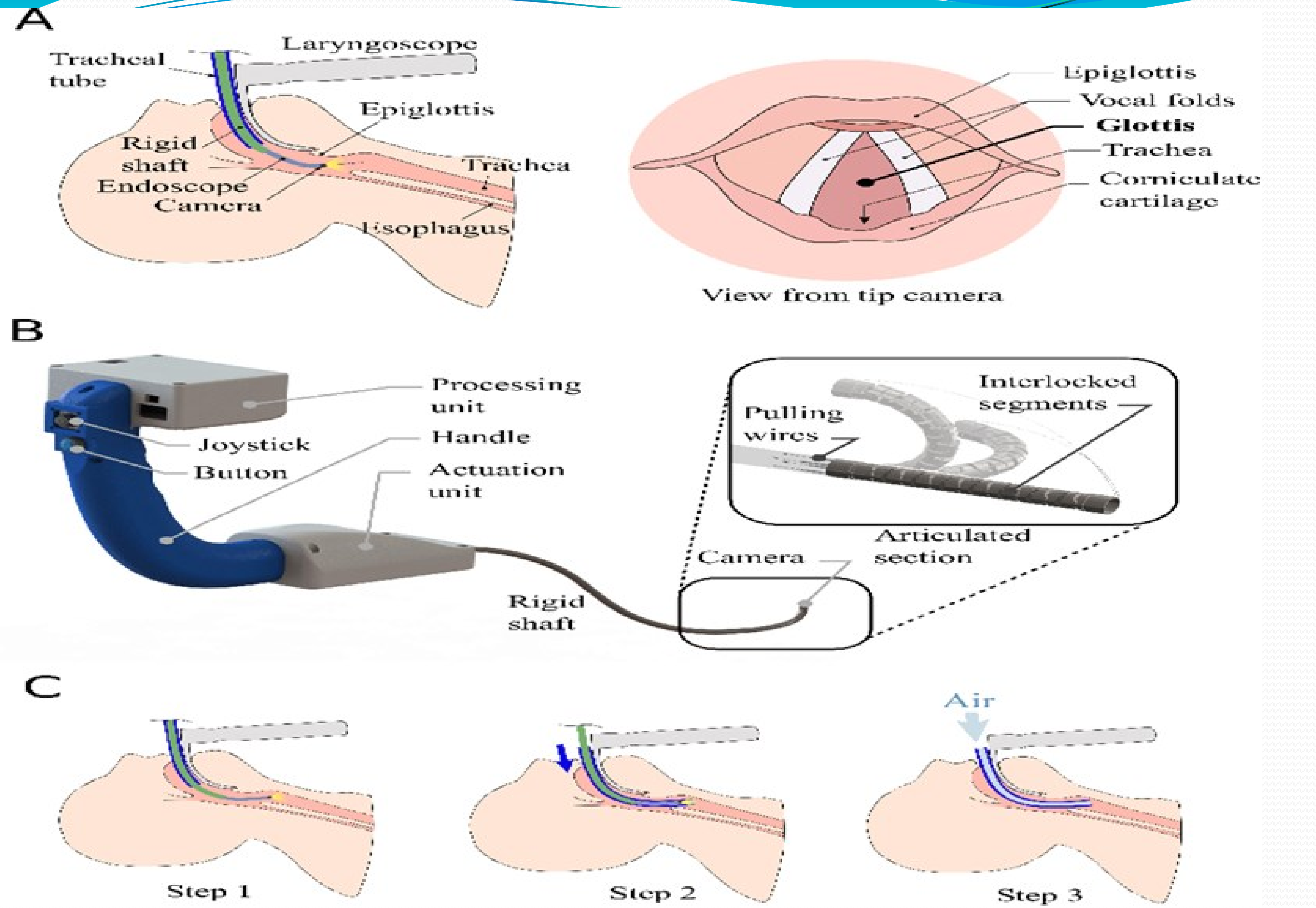
D. The vocal cords should be visualized prior to endotracheal placement.

- The glottic opening is recognized by its triangular shape and pale white vocal cords.
- Posteriorly, the vocal cords terminate in the arytenoids cartilages.
- The tube should be seen to pass between the cords, anterior to the

arytenoids.

- Insert the tube into the pharynx with the right hand from the right side of the mouth; it should pass without resistance through the vocal cords (about 1-2 cm).

The endotracheal tube cuff should lie in the upper trachea but beyond the larynx.


A
Laryngoscope
Tracheal tube
Epiglottis
Rigid shaft
Trachea
Endoscope
Camera
Esophagus
Epiglottis
Vocal folds
Glottis
Trachea
Corniculate cartilage
View from tip camera
B
Processing unit
Joystick
Handle
Button
Actuation unit
Interlocked segments
Pulling wires
Articulated section
Camera
Rigid shaft
C
Air
Step 1
Step 2
Step 3

Orotracheal Intubation

- ✓ The laryngoscope is held in the left hand. With the patient's mouth opened widely ,the blade is introduced into the right side of the oropharynx—with care to avoid the teeth.

- ✓ The tongue is swept to the left and up into the floor of the pharynx by the blade's flange.

- ✓ The tip of a curved blade is usually inserted into the vallecula, and the straight blade tip covers the epiglottis. With either blade,

- ✓ The handle is raised up and away from the patient in a plane perpendicular to the patient's mandible to expose the vocal cords ,

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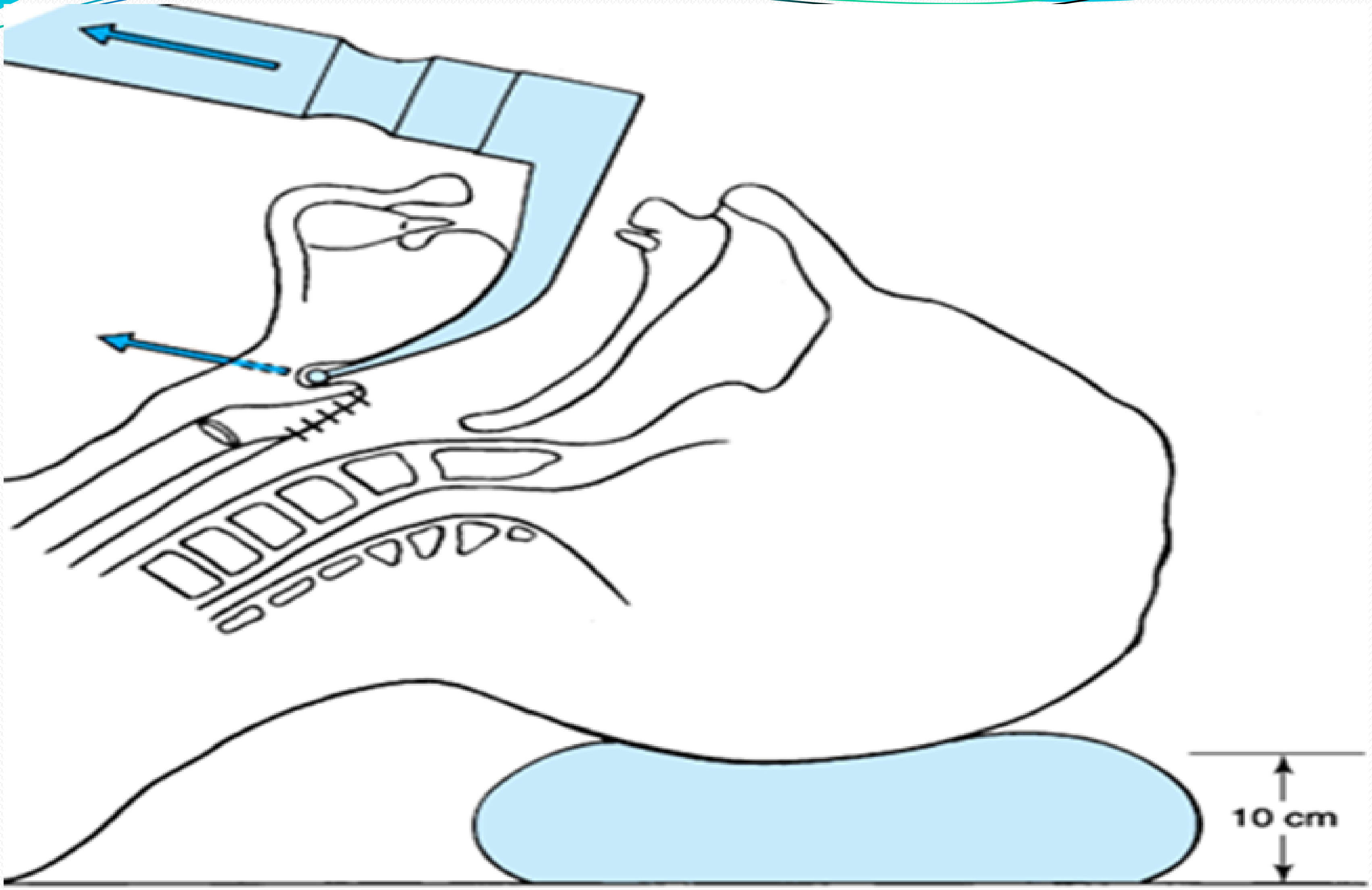
- ✓ Trapping a lip between the teeth and the blade and leverage on the teeth are avoided.
- ✓ The ETT is taken with the right hand, and its tip is passed through the abducted vocal cords.
- ✓ The ETT cuff should lie in the upper trachea but beyond the larynx.
- ✓ The laryngoscope is withdrawn, again with care to avoid tooth damage.
- ✓ The cuff is inflated with the least amount of air necessary to create a seal during positive-pressure ventilation to minimize the pressure transmitted to the tracheal mucosa.
- ✓ The cuff is inflated to seal the airway to deliver mechanical ventilation.
- ✓ A cuff pressure between 20 and 30 cm H2O is recommended to provide an adequate seal and reduce the risk of complications.
- ✓ Feeling the pilot balloon is *not a reliable method of* determining adequacy of cuff pressure


10 cm


alamy

Confirmation of endotracheal Intubation

A ,Direct visualization of the ET tube passing though the vocal cords.

B. Carbon dioxide in exhaled gases (documentation of end-tidal CO_2 in at least three consecutive breaths).

C. Bilateral breath sounds.

D. Absence of air movement during epigastric auscultation**.**

e. Condensation (fogging) of water vapor in the tube during exhalation.

F. Refilling of reservoir bag during exhalation.

G. Maintenance of arterial oxygenation.

H. Chest x-ray: the tip of ET tube should be between the carina and thoracic inlet or approximately at the level of the aortic notch or at the level of T5

Confirmation of endotracheal,,,,,,,,,,,

- ✓ After intubation, the chest and epigastrium are immediately auscultated and a capnographic tracing is monitored to ensure intra tracheal location

- ✓ If there is doubt about whether the tube is in the esophagus or trachea, it is prudent to remove the tube and ventilate the patient with a mask.

Otherwise, the tube is taped or tied to secure its position.

Although the persistent detection of CO_2 by a capnograph is the best confirmation of tracheal placement of a TT.

Confirmation of endotracheal,,,,,,,,,,,

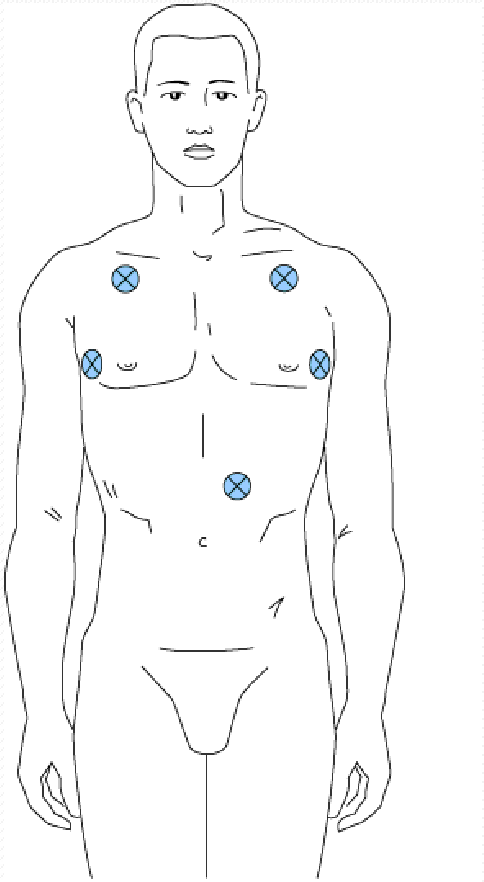
Proper tube location can be reconfirmed by

- Palpating the cuff in the sternal notch while compressing the pilot balloon with the other hand.

- The cuff should not be felt above the level of the cricoid cartilage, because a prolonged intralaryngeal location may result in postoperative hoarseness and increases the risk of accidental extubation.

- Tube position can be documented by chest radiography, but this is rarely required, except in an intensive care unit

Sites for auscultation of breath sounds at the apices and over the stomach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- A failed intubation should not be followed by repeated attempts that are merely more of the same.
- Changes must be made to increase the likelihood of success, such as repositioning the patient,
- Decreasing the tube size, adding a stylet, selecting a different blade, or requesting the assistance of another anesthetist
- If the patient is also difficult to ventilate with a mask, alternative forms of airway management (e.g. LMA, cricothyrotomy with jet ventilation, tracheostomy) must be immediately pursued.

Complications of endotracheal intubation

A.Complications occurring during intubation :aspiration, dental damage (chip tooth), laceration of the lips or gums, laryngeal injury, esophageal intubation, endobronchial intubation, activation of the sympathetic nervous system (high BP and HR), bronchospasm.

B. Complications occurring after extubation:aspiration, laryngospasm, transient vocalcord incompetence, glottic or subglottic edema, pharyngitis or tracheitis

Errors of Tracheal Tube Positioning

- Unintentional esophageal intubation can produce catastrophic results.
- Prevention of this complication depends on direct visualization of the tip of the TT passing through the vocal cords, careful auscultation for the presence of bilateral breath sounds and the absence of gastric gurgling while ventilating through the TT, analysis of exhaled gas for the presence of CO_2 (the most reliable method), chest radiography, or use of an FOB. Even though it is confirmed that the tube is in the trachea, it may not be correctly positioned.
- Over insertion usually results in intubation of the right main stem bronchus because of its less acute angle with the trachea.
- Clues to the diagnosis of bronchial intubation include unilateral breath sounds, unexpected hypoxia with pulse
- Oximetry (unreliable with high inspired oxygen concentrations), inability to palpate the TT cuff in the sternal notch during cuff inflation, and decreased breathing-bag compliance (high peak inspiratory pressures).

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- In contrast, inadequate insertion depth will position the cuff in the larynx, predisposing the patient to laryngeal trauma.

- Inadequate depth can be detected by palpating the cuff over the thyroid cartilage because no one technique protects against all possibilities for misplacing a ETT, minimal testing should include chest auscultation, routine capnography, and occasionally cuff palpation.

- If the patient is repositioned, tube placement must be reconfirmed.

- Neck extension or lateral rotation moves a ETT away from the carina, whereas neck flexion moves the tube toward the carina.

Tracheal Tube Malfunction

- ETTs do not always function as intended.
- The risk of polyvinyl chloride tube ignition in an environment was mentioned .
- Valve or cuff damage is not unusual and should be excluded prior to insertion.
- ETT obstruction can result from kinking, from foreign body aspiration, or from thick or inspissated secretions in the lumen.

Rapid sequence induction

- Patients who need general anesthesia, even though they have a full stomach (having recently eaten or having a condition that interferes with gastric emptying such as trauma or pregnancy), require especial technique, the so-called rapid sequence induction.

With a full stomach, the specter of regurgitation and aspiration arises. The predisposing factors for gastric aspiration include

a. Depressed level of consciousness

b. Impaired airway reflexes

c. Abnormal anatomical factors

d. Decreased gastro esophageal (GE) sphincter competence

e. Increased intragastric pressure

f. Delayed gastric emptying.

Steps in a rapid sequence induction

Once you have started a rapid sequence induction, you have lost the opportunity to check or obtain missing equipment. Thorough preparation therefore, is mandatory.

Preparation

1. ***Prepare and check for function:***

- suction
- intubation equipment
- tubes – one too large, one just right, one too small – check cuffs
- laryngoscope – two different blades – check lights
- machine
- emergency cricothyrotomy set available

2.**Have** available a helper skilled in applying cricoid pressure and to assist as necessary

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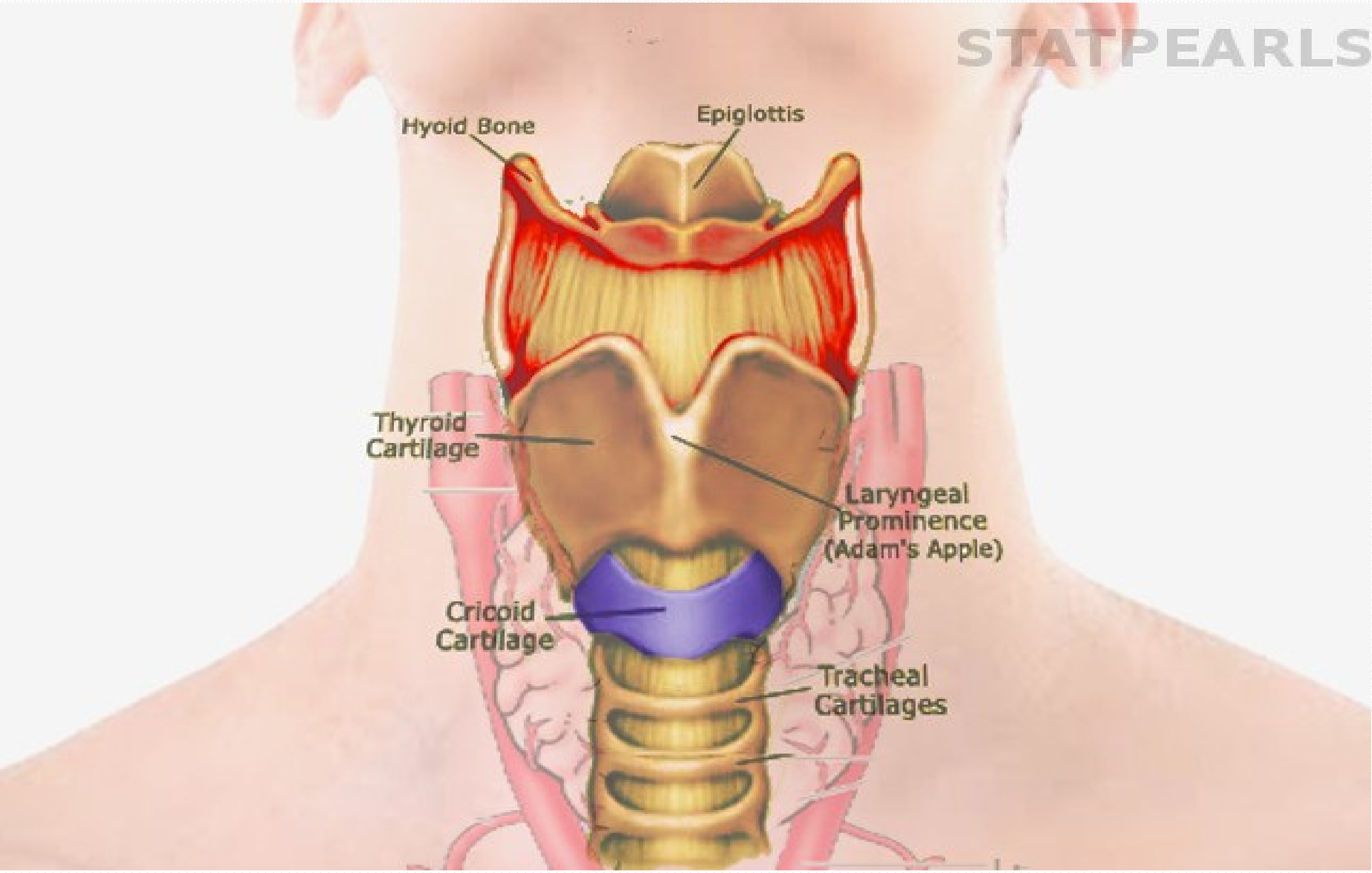

STATPEARLS
Hyoid Bone
Epiglottis
Thyroid Cartilage
Laryngeal Prominence (Adam's Apple)
Cricoid Cartilage
Tracheal Cartilages

…….

3. Prepare patient

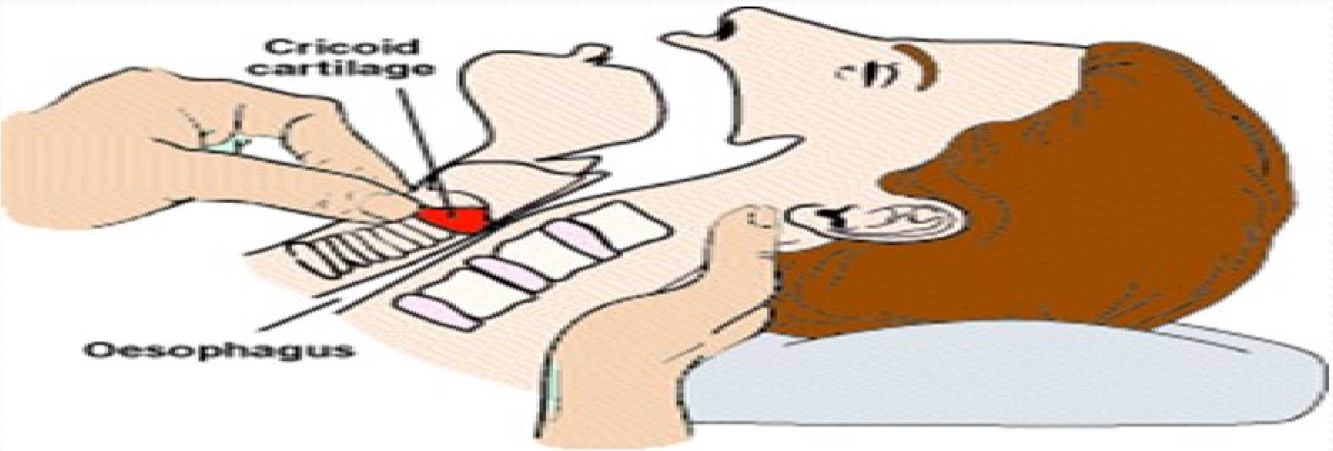
give antacid if circumstance permits obtain vital signs, print ECG strip

Induction

1. Pre-oxygenate/de-nitrogenate to an end-tidal oxygen of 80 to 90%
2. Tell the patient he will feel pressure on his neck as he falls asleep; meanwhile the assistant gently locates the cricoid ring
3. In rapid succession, administer an intubating dose of thiopental (or propofol) followed by an intubating dose of succinylcholine, while the assistant begins to apply cricoid pressure (10 newtons

CONT…….

4. As patient falls asleep, assistant increases cricoid pressure (30 newtons)



- Application of pressure to the cricoid cartilage to occlude the esophagus …..
- Sellick maneuver, is a technique used in endotracheal intubation to try to reduce the risk of regurgitation. .
- Thought to prevent the aspiration of gastric contents.
- VERY CONTROVERSIAL BUT STILL WIDELY PERFORMED

5.Sixty seconds after the succinylcholine entered the vein (or when apnea and relaxation coincide), intubate the trachea under direct laryngoscopy

6. Connect endotracheal tube to breathing circuit, inflate the cuff of the endotracheal tube then inflate the lung

7. Confirm endotracheal position of tube by watching chest rise – bilaterally

- listening for breath sounds – bilaterally in axillae
- listening over stomach for absence of breath sounds
- observing capnogram for appearance of carbon dioxide for 6 breaths.

8. Tell assistant to release cricoid pressure after confirming correct position of the tube

9. Secure tube and begin anesthesia

The goal of RSI is to rapidly secure and control the airway. Its all about TIME.

Positioning

- For many operations, the patient can lie on his back. Others require positions that may take an hour or more to be accomplished (for example, neurosurgical operations).

- We need to understand what position favors access for the surgeon and what positions present dangers for the patient (interference with ventilation ,compression of nerves, extreme flexion or extension of joints).
- Thus, the positioning is often a joint surgical/anesthesia task during which a lot of foam padding finds application between patient and hard surfaces.

- The most common post-operative nerve palsy affects the ulnar nerve which is exposed to pressure, being superficial and running through the ulnar groove at the elbow (between the medial epicondyle and the olecranon).

Maintenance

- Once we have placed the endotracheal tube or LMA and have confirmed its proper location by auscultation and end-tidal CO2, we can begin the administration of inhalation, intravenous (TIVA, total intravenous anesthesia) or a combination anesthetic.

- A number of halogenated drugs are available (halothane, isoflurane, desflurane, sevoflurane), but we use only one at a time

- Maintenance begins when the patient is sufficiently anesthetized to block awareness and movements in response to surgery.

- Vigilance on the part of the anesthetist is required to maintain homeostasis (vital signs, acid-base balance, temperature, coagulation, and volume status) and regulate anesthetic depth.

Depth of anesthesia and monitoring

- Once the patient is positioned, we must keep the anesthetic level so that the patient will neither feel pain nor remember the operation.

Yet this "anesthetic depth" must be balanced against the hemodynamic consequences (hypotension) of excess anesthetic, as well as the potential for delayed wake-up.

- If the patient is not paralyzed, there will be little doubt that he will move and let us know if he feels pain.
- We need to assess the depth of anesthesia clinically and with the help of instruments.
- The clinical assessment includes monitoring heart rate and blood pressure, which should be neither high from sympathetic response to noxious stimulation, nor low from overdose with anesthetics.,

MONITORING…………….

At the same time, we monitor

- pulse oximetry, blood pressure, heart rate, ECG, tidal volume, respiratory rate and peak inspiratory pressure, inspired oxygen, the concentration of respired gases and vapors, and the capnogram.

- blood loss, deep anesthesia, surgical activity (for example compressing the vena cava),an embolism (for example, air aspirated in an open vein……

CON,,,,,,

- we should be able to discover the effects as early as possible so that we can take corrective actions.

- We also assess the degree of muscle relaxation with the help of a nerve stimulator (twitch monitor) and by watching the operation and gauging muscle tone ,which might impede the surgeon's work.

- Thus we cannot be satisfied with watching the monitors; we need to keep an eye on the patient, his face, his position, and the surgeon's work.

DUCMENTATION

- A tedious aspect of our work is the obligation to keep a record of all these events and of our activities such as the administration of drugs and fluids, adjustment ventilator settings, and even of surgical events .

- Automated record keeping systems are becoming increasingly sophisticated.

Emergence from general anesthesia

- During this period, the patient makes the transition from an unconscious state to an awake state with intact protective reflexes.

Goals

- Patients should be awake and responsive, with full muscle strength and adequate pain control.

- Full recovery of airway reflexes and muscle function minimizes the risk of airway obstruction or pulmonary aspiration upon extubation and facilitates immediate neurologic assessment.

- In patients with cardiovascular disease, hemodynamic should be controlled

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- Well before the surgeon puts in the last stitch, we begin preparation for having the patient wake up.
- This might call for the reversal of a non-depolarizing neuromuscular blocking drug and the scaling back of inspired anesthetic concentrations
- Our goal is to have the patient awaken quickly and without pain; there fore, we titrate opioids or our regional anesthetic to anticipated the pain level without unacceptable respiratory depression, also considering the risk for postoperative nausea and vomiting.

Delayed awakening

- On occasion, a patient will not awaken promptly after the administration of general anesthesia due to;

A. ***Pharmacological*** cause depend on dose,absorption,.......

B. ***Metabolic cause,5H***

- ✓ Hypoglycemia,
- ✓ Hyperglycemia
- ✓ Hypernatremia plasma Na >145 ml/eq
- ✓ Hypokalemia,<3.5ml/eq
- ✓ Hypothermia

Uraemia,electrolyte imbalance,

C .***Respiratory cause*** primarly muscle problems,obisity....

D.***Neurological cause*** hypoxia or hemmorhage...

❖ Ventilator support and airway protection should be continued, and specific etiologies should be investigated.

Extubation

- Judging when to remove an ETT is part of the art of anesthesiology that develops with experience.
- It is an extremely important part of the practice as more complications arise during extubation and immediately afterward than with intubation. In general, extubation is best performed when a patient is either deeply anesthetized or awake.
- In either case, adequate recovery from neuromuscular blocking agents should be established prior to extubation

Equipment's for extubation

Supply of 100% oxygen
Face mask
Bag valve device with PEEP valve
End-tidal CO_2 detector
Suction equipment
Suction catheters
Large-bore tonsil suction apparatus (Yankauer)
Stylet
Magill forceps
Oral airways
Nasal airways
Laryngoscope handle and blades (curved, straight; various sizes)
Endotracheal tubes (various sizes)
Tongue depressors
Syringe for cuff inflation
Headrest
Supplies for vasoconstriction and local anesthesia
Tape
Tincture of benzoin

What do you need to extubate?

- Oxygen source
- Suction
- Oral/Nasal airways
- Face masks
- Endotracheal tubes
- LMA
- Pulse ox
- Cardiac Monitors
- CO_2 detectors
- Ambu bags

Criteria for extubation

The patient should be; Awake and hemodynamically stable. Have regained full muscle strength, be able to follow simple verbal commands (e.g., lift head), breathe spontaneously with acceptable oxygenation and ventilation.

Criteria	Suitability for Extubation
Hemodynamics	MAP, 50-80 mm Hg; heart rate, 50-90 beats/min; dobutamine <5 μg/kg; no sign of myocardial ischemia†
Ventilation	Spontaneous breathing; respiratory rate, 10-18/min; TV, ≥8 mL/kg
Consciousness	Obey simple commands
Muscular strength	Lift the head and hold for 30 s
Body temperature	37.0°C < 36.0°C
VAS	<3

*MAP indicates mean arterial pressure; TV, tidal volume; VAS, visual analogue scale.

†Sign of myocardial ischemia: change in S-T segment.

Technique

- The presence of an ETT may be irritating to patients emerging from anesthesia.

- Lidocaine (0.5 to 1.0 mg/kg IV) can be given to suppress coughing but may prolong emergence.

- The patient breathes 100% oxygen, and the oropharynx is suctioned.

- Mild positive airway pressure (20 cm H_2O) is applied via the ETT, the ETT cuff is deflated, and the tube is removed.

- Oxygen (100%) administration is continued by face mask.

- The anesthetist's attention should remain focused on the patient until the patient's ability to ventilate, oxygenate, and protect the airway is confirmed.

Deep extubation

- Stimulation of airway reflexes by the ETT during emergence can be avoided by extubation the trachea while the patient is still deeply anesthetized (stage III).

- This reduces the risk of laryngospasm and bronchospasm, making it a useful technique for severely asthmatic patients.

- It also avoids coughing and straining that may be undesirable after middle-ear surgery, open-eye procedures, and abdominal or inguinal herniorrhaphy

Awake extubation

- Extubation of the airway usually occurs after the patient fully regains protective reflexes.

- Awake extubation is indicated in patients at risk of aspiration of gastric contents, patients who have difficult airways, and patients who have just undergone tracheal or maxillofacial surgery.

- Before extubation, the patient should be awake and hemodynamically stable.

- The patient should have regained full muscle strength, be able to follow simple verbal commands (e.g., lift head), and breathe spontaneously with acceptable oxygenation and ventilation.

Problems

- Things don’t always run smoothly. If critical incidents occur, they must be discovered and corrected in time, lest they lead to disasters.

- To catch early trends ,however, presentsmore difficulties than one might think, because most signals we monitor are rather non-specific.

- Thus, a lowSpO_2 could be the result of malignant hyperthermia or faulty hospital piping, or low blood pressure the consequence of bleeding, deep anesthesia, or a measuring artifact.

- Therefore, with any deviation from normal, we need to think holistically about the patient and the anesthesia system with all of its components.

Patients may need re-intubation immediately or after some interval due to

- inappropriately timed extubation,
- progression of underlying disease,
- development of a new disorder.

Sometimes, in marginal patients extubation can be done with the expectation that the need for re-intubation is likely.

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